

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 13 (700 MHz) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0077
Revision:	Rev A.4
Author:	Ana Reyes
Reviewer:	Dr. Lena Kowalski
Design Center:	Irvine Design Center
Created Date:	2025-01-19
Last Modified:	2025-03-02
Status:	FAIL
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 13 (700 MHz) for UWB (Ultra-Wideband) Filter. The filter is designed on Langasite substrate with a target center frequency of 1718.5 MHz and bandwidth of 75.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 13 (700 MHz)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: Langasite
- Package: LGA (Land Grid Array)
- Design Tool: CST Studio Suite 2024
- Design Center: Irvine Design Center

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 13 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1718.5 MHz
Bandwidth (3 dB):	75.8 MHz
Insertion Loss Target:	≤ 1.8 dB
Return Loss Target:	≥ 14.0 dB
Out-of-Band Rejection:	≥ 30.0 dB
Isolation (Duplexer):	≥ 15.0 dB
Q Factor:	8016
Temperature Coefficient:	-25.5 ppm/°C
Die Size:	1.3 x 1.0 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 1718.5 MHz, BW 75.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 5156 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.8 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 114 MHz offset >= 30.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 5.5 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 90%

6. TEST RESULTS

Measurement Date: 2025-03-02
Devices Tested: 24
Insertion Loss (meas): 1.5 dB
Return Loss (meas): 15.8 dB
Rejection (meas): 33.2 dB
Isolation (meas): 20.2 dB
Q Factor (meas): 8016
Group Delay Variation: 5.3 ns
Power Handling: +35 dBm
Die Yield: 97.5%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated below-target performance for Band 13 (700 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

Author:	Ana Reyes	Date:	2025-03-02
Reviewer:	Dr. Lena Kowalski	Date:	2025-03-09
Approver:	Dr. James Liu	Date:	2025-03-12

END OF DOCUMENT - FD-TC-0077